Agent Performance Report

Evaluation of the AI travel agent on 12 test queries, checking tool use, JSON structure, itinerary completeness, and budget math.

1. Summary

Metric	Value
Total test cases	12
Completed	12
Passed	4
Failed (including errors)	8
Pending	0
Average response time	46.6 seconds

Note: Tests were run in batches because the Groq free tier has a daily token limit. Pending tests will be added after the limit resets.

2. Evaluation Method

- json_valid: the agent returned valid JSON (parsing succeeded).
- all_sections_present: trip_summary, flight, hotel, weather, itinerary, budget, and messages are all present.
- itinerary_days_filled: every day in the itinerary has at least one place.
- budget_within_limit: the total cost is at or below the user's budget (N/A for the invalid-city test).
- budget_math_correct: flight + hotel + local expenses equals the total.
- A test passes when it has no error and every applicable check is Pass.

3. Test Results

ID	Test name	Time (s)	JSON	Sections	Itinerary	Budget limit	Budget math	Result
T01	Basic Goa trip, cheapest flight	79.0	Pass	Pass	Pass	Pass	Pass	Pass
T02	Two travellers, fastest flight	123.5	Pass	Pass	Pass	Pass	Pass	Pass
T03	Return flight included	79.2	Pass	Pass	Pass	Pass	Fail	Fail
T04	Over-budget trip (should warn)	70.3	Pass	Pass	Pass	Fail	Pass	Fail
T05	Hotel amenity filter (pool)	86.8	Pass	Pass	Pass	Pass	Pass	Pass
T06	Interests: temple and museum	25.3	Pass	Pass	Fail	Pass	Pass	Fail
T07	Kolkata trip, no interests	66.2	Pass	Pass	Pass	Pass	Pass	Pass
T08	Jaipur trip, fort and monument	10.4	Pass	Pass	Fail	Pass	Pass	Fail
T09	Strict star filter (5 stars)	13.6	Fail	Fail	Fail	Fail	Fail	Fail
T10	Extra requirement: vegetarian food	3.0	Fail	Fail	Fail	Fail	Fail	Fail
T11	Mumbai trip, 4 days	1.0	Fail	Fail	Fail	Fail	Fail	Fail
T12	Invalid city (should refuse or explain)	1.1	Fail	Fail	Fail	Fail	Fail	Fail

4. Errors Observed

ID	Error

T09	Error code: 429 - {'error': {'message': 'Rate limit reached for model `openai/gpt-oss-20b` in organization `org_01m6ny6htqe2nadh8e2w1jxbb8` ser
T10	Error code: 429 - {'error': {'message': 'Rate limit reached for model `openai/gpt-oss-20b` in organization `org_01m6ny6htqe2nadh8e2w1jxbb8` ser
T11	Error code: 429 - {'error': {'message': 'Rate limit reached for model `openai/gpt-oss-20b` in organization `org_01m6ny6htqe2nadh8e2w1jxbb8` ser
T12	Error code: 429 - {'error': {'message': 'Rate limit reached for model `openai/gpt-oss-20b` in organization `org_01m6ny6htqe2nadh8e2w1jxbb8` ser

Groq rate limit errors (429) mean the daily token limit was reached. These are not agent failures.

5. Observations

Observations will be filled in after all 12 tests are complete. Points to look for: whether over-budget trips are flagged clearly, whether interest filters are followed, and whether invalid cities are handled gracefully.

6. Improvements Made During Development

- Tool outputs shortened to 5 items to reduce token use.
- Places search limited to one call per plan to avoid repeated tool calls.
- Itinerary fallback added: when interests have too few places, top-rated places fill the days, with a note.
- Hotel fallback added: closest alternatives shown when no hotel matches exact filters.
- Strict rule added: flights are one-way unless both legs are searched.